

Math 1232 Summer 2025

Mastery Quiz 10

Due Monday, August 4

This mastery quiz has three topics. Everyone should do M4, but if you already have a 4/4 on M3 or a 2/2 on S7, feel free to skip these topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Monday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Major Topic 4: Taylor Series
- Secondary Topic 7: Power Series

Name:

Name: _____

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2 + n - 3}{n^2 4^n}$.

(b) Analyze the convergence of the series $\sum_{n=1}^{\infty} n e^{-n^2+1}$.

(c) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n! + 1}{(n+1)!}$.

Major Topic 4: Taylor Series

(a) In class we computed a Taylor series for $\sin(x)$ centered at zero. Use the degree-seven Taylor polynomial to approximate $\sin(3) \approx T_7(3, 0)$. (You don't need to numerically simplify this.) Using the Taylor series remainder, find an upper bound for the error in this approximation.

(b) Using series we already know, write down a formula for the (infinite) Taylor series for $x^3 e^{(x^5/4)}$, and then write down the first four non-zero terms of this series.

(c) If $f(x) = \sum_{n=0}^{\infty} 2^n n^3 (x-2)^n$, compute $\frac{d}{dx} f(x)$ and $\int f(x) dx$.

Secondary Topic 7: Power Series

(a) Find the radius of convergence and the interval of convergence of $\sum_{n=0}^{\infty} (n(x-3))^n$.

(b) Find the radius of convergence and the interval of convergence of $\sum_{n=1}^{\infty} \frac{3^n (x+2)^n}{\ln(3n)}$.